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Report

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I'm working on a Pygame arm wrestling game for a lab. I've attached the code and README. Before changing anything, read it and tell me what the one deliberate bug is and why it happens.

The bug is in `handle_event`, in the two key branches:

```python
if event.key == pygame.K_LEFT:
    if self.last_key != pygame.K_LEFT:
        self.arm_position += 4.2
        ...
elif event.key == pygame.K_RIGHT:
    if self.last_key != pygame.K_RIGHT:
        self.arm_position += 4.2
        ...
```

Why it happens: the game uses `arm_position` as a signed scale. Negative values mean the player is winning (`arm_position <= -target_limit` sets `winner = "PLAYER"`), and positive values mean the computer is winning (`>= target_limit` sets `winner = "COMPUTER"`). The computer's force in `update()` also adds to `arm_position`, which is consistent with that convention.

But both of the player's alternating keypresses also do `arm_position += 4.2`. So every valid keystroke pushes the arm in the same direction as the computer, toward the computer's win threshold. The player is effectively helping the AI pin them, which matches the README's description of "pinning themselves."

The alternating-key logic itself is fine. The `last_key` check correctly requires Left, Right, Left, Right, and stamina drains correctly. The only problem is the sign of the change: it should be `-= 4.2` in both branches so the player's pushes move the arm toward `-target_limit`.

A couple of things I noticed while reading that aren't the bug but are worth knowing for the later tasks:

- **Rendering is consistent with the fix.** In `render`, `hand_x = width // 2 + offset_x` with the player on the left, so a negative `arm_position` moves the hands left, toward the player. That confirms the intended direction is negative.
- **Stamina gate:** `if self.stamina <= 10: return` blocks input at 10 or below, while the README says "falling below 10." That's a tiny off-by-one-ish mismatch, not part of the bug.
- **Per-frame balance:** the AI adds roughly 0.35 × (0.3 to 1.0), about 0.1 to 0.35 per frame, so once the sign is fixed the player needs to mash fairly consistently to make progress. You may want to retune after Task 2's surge phase.

I haven't changed anything yet. When you're ready, I can make the one-line sign fix and then we can move on to Task 2.

Fix the bug, then show me only the changed lines and explain the fix.
Then do Task 2 from the README: add an AI surge cycle. I want normal, surge (about 1.5s, much stronger), and cooldown (about 2.5s, weaker) phases that loop. Keep the timings as constants at the top of the class. Then Task 3. Add a flashing warning when the AI is surging, an "EXHAUSTED" message when the player's stamina is too low to push, and a bar showing the AI's current phase.Then Task 4. If the player pushes within the first second of the AI's cooldown, give a stamina boost and double push strength for about 3 seconds. Add an on-screen prompt for the timing window and show the boost while it's active. Give me only the changes i have to make to the code

Step 1: Fix the bug

In `handle_event`, change `+=` to `-=` on both branches (the Left and Right ones):

diff

```
            if event.key == pygame.K_LEFT:
                if self.last_key != pygame.K_LEFT:
-                   self.arm_position += 4.2
+                   self.arm_position -= 4.2
...
            elif event.key == pygame.K_RIGHT:
                if self.last_key != pygame.K_RIGHT:
-                   self.arm_position += 4.2
+                   self.arm_position -= 4.2
```

Why: `arm_position` is signed. Negative moves toward the player's win at `-target_limit`, and positive moves toward the computer's win. The player's pushes were adding to it, so they were helping the computer.

Step 2: AI surge cycle

Add at the top of the class, above `__init__`:

```python
class GameEngine:
    # AI phase durations (milliseconds)
    AI_NORMAL_MS = 4000
    AI_SURGE_MS = 1500
    AI_COOLDOWN_MS = 2500

    # AI push per frame (multiplied by a 0.3-1.0 random variance in update)
    AI_NORMAL_FORCE = 0.35
    AI_SURGE_FORCE = 1.0
    AI_COOLDOWN_FORCE = 0.12

    PHASE_MS = {"NORMAL": AI_NORMAL_MS, "SURGE": AI_SURGE_MS, "COOLDOWN": AI_COOLDOWN_M
    PHASE_FORCE = {"NORMAL": AI_NORMAL_FORCE, "SURGE": AI_SURGE_FORCE, "COOLDOWN": AI_C
    NEXT_PHASE = {"NORMAL": "SURGE", "SURGE": "COOLDOWN", "COOLDOWN": "NORMAL"}
```

In `__init__`, delete `self.ai_strength = 0.35` and add:

```python
        self.ai_phase = "NORMAL"
        self.phase_start = pygame.time.get_ticks()
```

In `update()`, replace the two AI force lines:

```diff
-         ai_variance = random.uniform(0.3, 1.0)
-         self.arm_position += self.ai_strength * ai_variance
+         now = pygame.time.get_ticks()
+         self._update_ai_phase(now)
+
+         ai_variance = random.uniform(0.3, 1.0)
+         self.arm_position += self.PHASE_FORCE[self.ai_phase] * ai_variance
```

In `reset()`, add:

```python
        self.ai_phase = "NORMAL"
        self.phase_start = pygame.time.get_ticks()
```

New method:

```python
```

```
    def _update_ai_phase(self, now):
        if now - self.phase_start >= self.PHASE_MS[self.ai_phase]:
            self.ai_phase = self.NEXT_PHASE[self.ai_phase]
            self.phase_start = now
```

Step 3: Warnings, exhaustion, and phase bar

Add to the class constants:

python

```
    EXHAUSTED_THRESHOLD = 10
```

In `__init__`, add a small font:

python

```
        self.font_small = pygame.font.SysFont(None, 22)
```

In `handle_event`, use the constant:

diff

```
-           if self.stamina <= 10:
+           if self.stamina <= self.EXHAUSTED_THRESHOLD:
                return
```

In `render()`, add this line just before `if self.game_state == "GAME_OVER":`:

python

```
        self._render_indicators(screen)
```

New method:

python

```
    def _render_indicators(self, screen):
        now = pygame.time.get_ticks()
        cx = self.width // 2
        blink = (now // 150) % 2 == 0

        # AI phase bar (drains as the phase runs out)
        colors = {"NORMAL": (90, 140, 220), "SURGE": (230, 60, 60), "COOLDOWN": (70, 20
        bar = pygame.Rect(cx - 130, 62, 260, 24)
        remaining = max(0.0, 1.0 - (now - self.phase_start) / self.PHASE_MS[self.ai_pha
        pygame.draw.rect(screen, (45, 50, 60), bar, border_radius=6)
        fill = pygame.Rect(bar.x, bar.y, int(bar.width * remaining), bar.height)
        pygame.draw.rect(screen, colors[self.ai_phase], fill, border_radius=6)
        pygame.draw.rect(screen, (200, 200, 200), bar, width=2, border_radius=6)
        label = self.font_small.render(f"AI: {self.ai_phase}", True, (255, 255, 255))
        screen.blit(label, (cx - label.get_width() // 2, bar.centery - label.get_height

        # Flashing surge warning
        if self.ai_phase == "SURGE" and blink:
            pygame.draw.rect(screen, (255, 60, 60),
                             pygame.Rect(40, 100, self.width - 80, 310),
                             width=6, border_radius=14)
            warn = self.font_big.render("!! AI SURGE !!", True, (255, 80, 80))
            screen.blit(warn, (cx - warn.get_width() // 2, 112))

        # Exhaustion message next to the stamina bar
        if self.stamina <= self.EXHAUSTED_THRESHOLD:
            color = (255, 60, 60) if blink else (150, 40, 40)
            msg = self.font_med.render("EXHAUSTED", True, color)
            screen.blit(msg, (395, 447))
```

Step 4: Counter-surge

Add to the class constants:

python

```
    PUSH_STRENGTH = 4.2
    STAMINA_COST = 2.0
    COUNTER_WINDOW_MS = 1000
    BOOST_DURATION_MS = 3000
    BOOST_STAMINA = 40.0
    BOOST_MULTIPLIER = 2.0
```

In `__init__` and `reset()`, add:

python

```
        self.counter_used = False
        self.boost_until = 0
```

In `_update_ai_phase`, reset the one-shot flag when cooldown begins:

diff

```
             self.ai_phase = self.NEXT_PHASE[self.ai_phase]
             self.phase_start = now
+            if self.ai_phase == "COOLDOWN":
+                self.counter_used = False
```

In `handle_event`, replace the whole `if event.type == pygame.KEYDOWN:` block with the version below. It supersedes the two branches from Step 1 and keeps the fix, because `_push` subtracts:

python

```
        if event.type == pygame.KEYDOWN:
            if self.stamina <= self.EXHAUSTED_THRESHOLD:
                return
            if event.key in (pygame.K_LEFT, pygame.K_RIGHT) and event.key != self.last_
                self._push(event.key)
```

New methods:

python

```
    def _push(self, key):
        now = pygame.time.get_ticks()
        self._try_counter_surge(now)
        strength = self.PUSH_STRENGTH * (self.BOOST_MULTIPLIER if self.boost_active(now
        self.arm_position -= strength
        self.stamina = max(0.0, self.stamina - self.STAMINA_COST)
        self.last_key = key

    def _try_counter_surge(self, now):
        if (self.ai_phase == "COOLDOWN"
                and not self.counter_used
                and now - self.phase_start <= self.COUNTER_WINDOW_MS):
            self.counter_used = True
            self.boost_until = now + self.BOOST_DURATION_MS
            self.stamina = min(self.max_stamina, self.stamina + self.BOOST_STAMINA)

    def boost_active(self, now):
        return now < self.boost_until
```

Append to the end of `_render_indicators` (same indentation as the rest of the method):

```python
        # Counter-surge prompt / active boost
        if self.boost_active(now):
            secs = (self.boost_until - now) / 1000
            txt = self.font_med.render(
                f"COUNTER-SURGE x{int(self.BOOST_MULTIPLIER)}  {secs:.1f}s", True, (255
            screen.blit(txt, (cx - txt.get_width() // 2, 155))
        elif (self.ai_phase == "COOLDOWN" and not self.counter_used
              and now - self.phase_start <= self.COUNTER_WINDOW_MS):
            txt = self.font_big.render("PUSH NOW!", True, (255, 220, 60))
            screen.blit(txt, (cx - txt.get_width() // 2, 155))
```

Notes

- **Exhaustion barely triggers right now.** Stamina regenerates 0.8 per frame (about 48 per second at 60 FPS) against a cost of 2 per push, so a human can't drain it. For the "after" video, change the regen line in `update()` to something like `+ 0.15` and the "EXHAUSTED" state will show up.
- **Untested:** I haven't run this, and I haven't seen `main.py`, so the layout assumes the window is at least about 520px wide. If the HUD overlaps anything, nudge the coordinates.
- **Balance:** all force values and timings are constants at the top of the class, so you can tune them there.

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Review the full file against the README's expected behavior and tell me if anything is wrong or missing. Also, is the exhaustion state actually reachable with the current stamina numbers?

Short answer on exhaustion: no, it's not reachable. Stamina is effectively non-functional right now.

- Each valid push costs 2.0 stamina, and regen is 0.8 per frame, so at 60 FPS that's about 48 stamina per second.
- To drain stamina at all, the player would need more than 24 valid presses per second. Realistic alternating mashing is about 8-12 per second, so drain is 16-24 per second against 48 per second of regen.

- The bar just dips 2 points and refills within a few frames. It sits pinned near full, and "EXHAUSTED!" can never display.
- This also guts Task 4. The "stamina recovery boost" refunds stamina when the player is already near 100, so it does nothing. Only the double push strength has any effect.

Fix

Pull the magic numbers into constants and retune them:

```python
    STAMINA_COST = 3.0
    BASE_REGEN = 0.15           # ~9/sec
    COUNTER_EXTRA_REGEN = 0.6   # was 1.5; ~45/sec total while boosted
```

In `handle_event`:

```diff
-            self.stamina = max(0.0, self.stamina - 2.0)
+            self.stamina = max(0.0, self.stamina - self.STAMINA_COST)
```

In `update()`:

```diff
-        regen = 0.8
+        regen = self.BASE_REGEN
```

At 10 presses per second, drain is about 30 per second against about 9 per second of regen, so sustained mashing exhausts the player in roughly 4-5 seconds. During the counter-surge, regen (about 45 per second) beats drain, so the boost does real work. Tune the numbers to taste.

Other issues

1. The exhaustion state will flicker once it's reachable. With `<= 10`, the player recovers to 10.1, pushes once, drops to 7.1, and is locked again, so the message strobes. A resume threshold fixes it (optional):

```python
    EXHAUST_RESUME = 25
```

In `__init__` and `reset()`:

```python
        self.exhausted = False
```

Replace the helper:

```python
    def is_exhausted(self):
        return self.exhausted
```

In `update()`, right after the stamina regen block:

```python
        if self.stamina <= self.EXHAUST_THRESHOLD:
            self.exhausted = True
        elif self.stamina >= self.EXHAUST_RESUME:
            self.exhausted = False
```

2. `counter_flash_timer` is dead code. It's set to 60 and decremented but never drawn. Either delete it or render it. This replaces the "COUNTER NOW" prompt at the same spot, since `in_counter_window()` goes false once the counter fires. Add it in `render()` before the counter prompts:

```python
        if self.counter_flash_timer > 0:
            flash = self.font_big.render("COUNTER!", True, (255, 255, 255))
            screen.blit(flash, (self.width // 2 - flash.get_width() // 2, 530))
```

3. The "COUNTER NOW! PUSH!" prompt shows while the player is exhausted and can't push, and it draws at the same y as "EXHAUSTED!" (530). Change the condition:

```diff
-        if self.in_counter_window() and flash_on:
+        if self.in_counter_window() and not self.is_exhausted() and flash_on:
```

4. Check `main.py` for window height and FPS. I haven't seen it, and two things depend on it:

- The new HUD runs down to about y=600 (AI bar at 488, indicators at 530 and 585). The original layout ended around y=470, so if `main.py` creates a window shorter than about 620px, the warnings will be off-screen.
- All the timers are in frames and assume `clock.tick(60)`. At a different FPS, "1.5s surge" will be wrong.

5. Nits:

- `import math` is unused.
- The README says input halts when stamina falls "below 10", but the code uses `<= 10`. This is harmless.

- The AI POWER bar uses fixed fill levels per phase (full, 50%, 20%) rather than time remaining. That's valid for a "power level" bar, but a draining countdown would make the 1-second counter window easier to read.

What checks out against the README

- Task 1: `arm_position -= push` fixes the inversion, and the signed convention matches the win conditions and the render.
- Task 2: the NORMAL → SURGE → COOLDOWN loop works, and the timings and multipliers are constants.
- Task 3: the flashing surge warning and the phase bar are present.
- Task 4: the one-shot per cooldown is handled by `counter_used_this_cooldown`, and the boost (180 frames) ends during the following normal phase, so it never overlaps a surge.
- Win detection at ±100 and `reset()` clearing all the new state are correct.

I haven't run the game, so confirm the retuned stamina feel by playing it. For your "after" video, a few seconds of nonstop mashing should now show the stamina bar draining and "EXHAUSTED!" appearing.